

CH.1 - Variations in Psychological Attributes

One of the main concerns of modern psychology has been the study of individual differences.

• Individual Differences in Human Functioning

- For psychologists, individual differences refer to distinctiveness and variations among people's characteristics and behaviour patterns.
- Some psychologists believe that our behaviours are influenced by our personal traits, while others hold the view of situationism. Situationism states that situations and circumstances in which one is placed, influence one's behaviour.
eg. An aggressive individual may behave submissively in the presence of their boss.

• Assessment of Psychological Attributes

- Assessment is the first step in understanding a psychological attribute.
- Psychological Assessment uses systematic testing procedures to evaluate abilities, behaviours, and personal qualities of individuals.
- Assessment can be informal or formal
 - * INFORMAL ASSESSMENT - varies from case-to-case & from one assessor to another, open to subjective interpretations.
 - * FORMAL ASSESSMENT - objective, standardised, and organised
- Psychologists are trained in making formal assessment of psychological attributes.
- Once assessment is done, we can use the information to predict.
- The attribute chosen for assessment depends upon our purpose.

• Some Domains of Psychological Attributes

- Psychological attributes are complex and multi-dimensional.
eg. if you want to have a complete assessment of a person, you will need to assess their functioning in various domains or areas.

Attributes that are of interest to psychologists :

1. INTELLIGENCE

It refers to the global capacity to understand the world, think rationally, and use available resources effectively when faced with challenges.

Intelligence tests provide a global measure of a person's general cognitive competence.

2. APTITUDE

It refers to an individual's underlying potential for acquiring skills.

Aptitude tests are used to predict what an individual will be able to do if given a proper environment and training.

3. INTEREST

It refers to an individual's preference for engaging in one or more specific activities relative to others.

Knowledge of interests helps us in making choices that promote life satisfaction and performance on jobs.

4. PERSONALITY

It refers to relatively enduring characteristics of a person that make her/him distinct from others.

Personality tests try to assess an individual's unique characteristics, eg. moody or emotionally stable, outgoing or withdrawn, etc.

5. VALUES

Values are enduring beliefs about an ideal model of behaviour. A person having a value sets a standard of guiding their actions in life and also for judging others.

• Assessment Methods

① Psychological Test is an objective and standardised measure of an individual's mental and/or behavioural characteristics.

These tests are widely used for the purposes of clinical diagnosis, guidance, personnel selection, placement & training. Tests could be objective or projective.

② Interview involves seeking information from a person on a one-to-one basis. eg. counsellor interacting with a client, journalists interviews, employee selection interviews, etc.---

③ Case Study is an in-depth study of the individual in terms of their psychological attributes, psychological history in the context of her/his psychosocial and physical environment. Case studies are based on data generated by different methods like observation, interview, questionnaire, etc.---

(4) Observation involves employing systematic, organised, and objective procedures to record behavioural phenomena occurring naturally in real time.
eg. studying mother-child interaction

The major problems with observational methods are that the observer has little control over the situation & reports may suffer from subjective interpretations of the observer.

(5) Self-Report is a method in which a person provides factual information about herself/himself and/or opinions, beliefs, etc... that s/he holds.

• INTELLIGENCE

- Intelligence provides an understanding of how people adapt their behaviour according to the environment they live in.
- Oxford Dictionary explains intelligence as the power of perceiving, learning, understanding, and knowing.
- Psychological notion of intelligence is quite different from the common sensical notion of intelligence.

Early Intelligence Theorists :

a) Alfred Binet

- One of the first psychologists who worked on intelligence.
- He defined intelligence as the ability to judge well, understand well, and reason well.

(b) Wechsler

- Wechsler's intelligence tests are most widely used
- He defined intelligence as the global and aggregate capacity of an individual to think rationally, act purposefully, and to deal effectively with her/his environment.

(c) Gardner & Sternberg

- Gardner & Sternberg have suggested that an intelligent individual not only adapts to the environment, but also actively modifies or shapes it.

THEORIES OF INTELLIGENCE

Psychometric Approach

- Binet's Theory
- Charles Spearman's two factor theory
- Louise Thurstone's
- Arthur Jensen
- J.P. Guilford

Information-Processing Approach

- Theory of Multiple Intelligences
- Triarchic Theory
- PASS Model of Intelligence

(I) PSYCHOMETRIC APPROACH

This approach considers intelligence as an aggregate of abilities. It expresses an individual's performance in terms of a single index of cognitive abilities. Major focus is on structure of intelligence.

LET'S STUDY THE THEORIES BASED ON THIS APPROACH !

(a) Binet's Theory of Intelligence

- This theory was simple as it arose from his interest in differentiating more intelligent from less intelligent individuals.
- He conceptualised intelligence as consisting of one similar set of abilities which can be used for solving any or every problem in an individual's environment.
- His theory is called Uni or One-Factor Theory of intelligence.
- However, this theory came to be disputed.

(b) Spearman's Two-Factor Theory

- By Charles Spearman, 1927
- He employed a statistical method called factor analysis and showed that intelligence consisted of a general factor (g-factor) and some specific factors (s-factor).
- g-factor includes mental operations which are primary & common to all performances.
- s-factor includes specific abilities which allow people to excel in their respective domains.
eg. excellent singers, architects, scientists, athletes.

(c) Thurstone's Theory of Primary Mental Abilities

- Proposed by Louis Thurstone
- It states that intelligence consists of 7 primary abilities, each of which is relatively independent of the others.

The abilities are :

- i) Verbal Comprehension (grasping meaning of words, concepts & ideas)
- ii) Numerical Abilities (speed & accuracy in numerical & computational skills)
- iii) Spatial Relations (visualising patterns and forms)
- iv) Perceptual Speed (speed in perceiving details)
- v) Word Fluency (using words fluently & flexibly)
- vi) Memory (accuracy in recalling information)
- vii) Inductive Reasoning (deriving general rules from presented facts).

(d) Arthur Jensen's Hierarchical Model of Intelligence

- This model consists of abilities operating at 2 levels, called Level I and Level II.
- Level I is the associative learning in which output is more or less similar to the input.
(eg. rote learning and memory)
- Level II, called cognitive competence, involves higher-order skills as they transform the input to produce an effective output.

(e) J. P. Guilford's Structure-of-Intellect Model

- Introduced by J. P. Guilford, 1988
- It classifies intellectual traits among 3 dimensions:
 - 1) Operations are what the respondent does.
These include, ~~memory~~, cognition, memory retention, divergent production, convergent production, memory recording & evaluation.

- 2) Contents refers to the nature of materials or information on which intellectual operations are

performed. These include visual, auditory, symbolic (eg, letters, numbers), semantic (eg, words), and behavioural (eg, information about people's behaviour, attitudes, and needs)

3) Products refer to the form in which information is processed by the respondent.

Products are classified into units, and classes, relations, systems, transformations, and implications.

- Guilford's Classification includes $6 \times 5 \times 6$ categories, thus, the model has 180 cells. Each cell is expected to have at least one factor or ability; some cells may have more than one factor. Each factor is described in terms of all 3 dimensions.

II INFORMATION - PROCESSING APPROACH

This approach describes the processes people use in intellectual reasoning and problem solving. The major focus of this approach is on how an intelligent person acts.

(a) Theory of Multiple Intelligence

- Proposed by Howard Gardner
- According to him, intelligence is not a single entity; rather distinct types of intelligences exist. Each of these are independent of each other.
- Gardner studied extremely talented persons, who had shown exceptional abilities in their respective areas, and described 8 Types of intelligence:

- ① Linguistic (skills involved in the production & use of language)
- It is the capacity to use language fluently and flexibly to express one's thinking and understanding topics & others.
 - Persons high on this intelligence are 'word-smart', i.e. they are sensitive to different shades of word meanings, can create linguistic images in their mind.
- eg. Poets & writers

② Logical-Mathematical (skills involved in scientific thinking and problem solving)

- People high on this type of intelligence can think logically and critically. They engage in abstract reasoning, and can manipulate symbols to solve mathematical problems.
- eg. Scientists & Nobel Prize winners

③ Spatial (skills in forming visual images & patterns)

- Refers to the abilities involved in forming, using, and transforming mental images.
- People high on this can easily represent the spatial world in mind.

eg. Pilots, sailors, sculptors, painters, architects, doctors & interior decorators.

④ Musical (sensitivity to musical rhythms & patterns)

- It is the capacity to produce, create, and manipulate musical patterns.
- People high on this intelligence are very sensitive to sounds and vibrations, and in creating new patterns of sounds.

⑤ Bodily-Kinaesthetic (using whole or parts of the body flexibly and creatively)

- This consists of the use of the whole body or parts of it for display or construction of products and problem solving.

eg. Athletes, dancers, actors, sports persons, gymnasts, & surgeons.

⑥ Interpersonal (sensitivity to subtle aspects of others' behaviour)

- This is the skill of understanding the motives, feelings and behaviour of other people so as to bond into a comfortable relationship with others.

eg. Psychologists, counsellors, politicians, social workers, and religious leaders.

⑦ Intrapersonal (awareness of one's own feelings, motives, and desires)

- This refers to the knowledge of one's internal strengths and limitations & using that knowledge to effectively relate to others.

eg. Philosophers & spiritual leaders.

⑧ Naturalistic (sensitivity to the features of the natural world)

- This involves complete awareness of our relationship with the natural world. It's useful in recognizing the beauty of different species, of flora and fauna.

eg. Hunters, farmers, tourists, botanists, zoologists.

(b) Triarchic Theory of Intelligence

- Robert Sternberg (1985)
- Sternberg views intelligence as "the ability to adapt, to shape, and select environment to accomplish one's goals and those of one's society and culture".
- Acc. to this theory, there are 3 Basic Types of Intelligence

(i) Componential Intelligence

- Componential/Analytical Intelligence is the analysis of information to solve problems.
- People high on this ability think analytically, critically & succeed in schools.
- This intelligence has 3 components :
 - * Knowledge Acquisition component, responsible for learning & acquisition of the ways of doing things
 - * Meta or a Higher Order component, which involves planning concerning what to do and how to do.
 - * Performance component, which involves actually doing things.

(ii) Experiential Intelligence

- Experiential/Creative Intelligence is involved in using past experiences creatively to solve novel problems. It is reflected in creative performance.
- People high on this aspect integrate different experiences in an original way to make new discoveries & inventions. They quickly find out which info. is useful in a given situation.

(iii) Contextual Intelligence

- Contextual/Practical intelligence involves the ability to deal with environmental demands encountered on a daily basis.
- It maybe called "street smarts" or "business sense".
- People high on this aspect easily adapt to their present environment or select a more favorable environment than the existing one, or modify the environment to fit their needs.
- ° They turn out to be successful in life.

(c) Planning, Attention - arousal, and Simultaneous - successive (PASS) Model of Intelligence

- Developed by J.P. Das, Jack Naglieri, & Kirby (1994)
- Acc. to this model, intellectual activity involves the interdependent functioning of 3 neurological systems called functional units of brain.
- These units are responsible for arousal/attention, coding or processing, and planning respectively.

(i) Arousal/Attention

- State of arousal is basic to any behaviour as it helps us in attending to stimuli.
- Arousal & attention enable a person to process information.
- Too much or too little arousal would interfere with attention. ° Optimum level of arousal is required.

(ii) Simultaneous and Successive Processing

- The information can be integrated into your knowledge system either simultaneously or successively.

- Simultaneous Processing takes place when you perceive the relations among various concepts & integrate them into a meaningful pattern for comprehension.
- Successive Processing takes place when you remember all the information serially so that the recall of one leads to the recall of another. eg. digits, alphabets, multiplication tables...

(iii) Planning

- It allows us to think of the possible courses of action, implement them to reach a target, and evaluate their effectiveness.
- If a plan does not work, it is modified to suit the requirements of the task or situation.

These PASS Processes operate on a knowledge base developed either formally (by reading, writing, & experimenting) or informally from the environment.

⇒ Das & Naglieri developed a battery of tests, known as the Cognitive Assessment System (CAS).

It consists of verbal as well as non-verbal tasks that measure basic cognitive functions presumed to be independent of schooling.

Meant for individuals between 5-18 years of age.

• Individual Differences in Intelligence

Intelligence : Interplay of Nature & Nurture

(a) Role of Heredity (NATURE)

- The evidence for hereditary influences on intelligence comes mainly from studies on twins & adopted children.
- The intelligence of identical twins reared together correlate almost 0.90
- Intelligence of identical twins reared in different environments correlate 0.72
- Fraternal twins reared together $\rightarrow$ 0.60
- Siblings reared together $\rightarrow$ 0.50
- Siblings reared apart $\rightarrow$ 0.25
- Studies of adopted children, shows that children's environment intelligence is more similar to their biological parents rather than adoptive parents.

(b) Role of Environment (NURTURE)

- Studies have reported that as children grow in age, their intelligence levels tend to move closer to that of their adoptive parents.
- There is evidence that environmental deprivation lowers intelligence while rich nutrition, good family background, and quality schooling increases intelligence.

Thus, Intelligence is a product of complex interaction of Heredity (Nature) and Environment (Nurture).

Assessment of Intelligence

- In 1905, Alfred Binet & Theodore Simon, made the first successful attempt to formally measure intelligence.
- In 1908, they gave the concept of Mental Age (MA), which is a measure of a person's intellectual development relative to people of her/his age group.
- Chronological Age (CA) is the biological age from birth.
 - A bright child's MA is more than her/his CA
 - For a dull child, MA is below CA.
 - They described 'intellectual disability' as being 2 mental ages years below the chronological age.
- In 1912, William Stern (German Psychologist) gave the concept of IQ (Intelligence Quotient).

$$IQ = \frac{MA (Mental Age)}{CA (Chronological Age)} \times 100$$

- (The number '100' is used as a multiplier to avoid the decimal)
- # The average IQ in the population is 100, irrespective of age.
 - # The frequency distribution of the IQ scores tends to approximate a bell-shaped curve, called "Normal Curve".

<u>IQ Range</u>	<u>Descriptive Label</u>
Above 130	Very Superior
120 - 130	Superior
110 - 119	High average
90 - 109	Average
80 - 89	Low Average
70 - 79	Borderline
Below 70	Mentally Challenged

Intellectually gifted individuals $\rightarrow$ IQ above 130
Mentally challenged individuals $\rightarrow$ IQ below 70

(These 2 groups deviate considerably from the normal population in respect of their cognitive, emotional, and motivational characteristics).

Variations of Intelligence

① Intellectual Deficiency

- Those children who show intellectual deficiency are termed as 'mentally challenged' or 'mentally retarded'.

- American Association on Mental Deficiency (AAMD)

views mental retardation as "significantly sub-average general intellectual functioning existing concurrently with deficits in adaptive behaviour and manifested during the developmental period".

① First, in order to be judged as mentally retarded, a person must show significantly sub-average intellectual functioning. (i.e., IQ below 70)

② Second, relates to deficits in adaptive behaviour.

Adaptive behaviour refers to a person's capacity to be independent & deal effectively with one's environment.

③ Third is that the deficits must be observed during the developmental period (i.e., btw. 0-18 years).

- Different levels of retardation :

• Mild Retardation (IQs 55-69), they can function quite independently, hold jobs, and families, although their development is typically slower than their peers.

- Moderate Retardation (IQs 40-54) - lag behind their peers in language & motor skills. They need to have moderate degree of supervision in everyday tasks.

- Severe Retardation (IQs 25-39) & Profound Retardation (IQs below 25)

- individuals are incapable of managing life and need constant care for their entire lives.

② Intellectual Giftedness

- Intellectually gifted individuals show higher performance because of their outstanding potentialities.
- Study of gifted individuals began in 1925, by Lewis Terman.
 - Giftedness is exceptional general ability shown in superior performance in a wide variety of areas.
 - Talent is a narrower term & refers to remarkable ability in a specific field. Highly talented are sometimes called "prodigies"
- Giftedness from teachers' point of view depends on a combination of 'high ability', 'high creativity', & 'high commitment'.
- Early Signs of intellectual superiority (i.e. giftedness): during infancy & early childhood children show
 - larger attention span
 - good recognition memory
 - preference for novelty
 - sensitivity to environmental changes
 - early appearance of language skills

Important Characteristics of Gifted Children :

- Advanced logical thinking, questioning, & problem-solving behaviour,
- High speed in processing information,
- Superior generalisations and discrimination ability,
- Advanced level of original & creative thinking,
- High level of intrinsic motivation & self-esteem,
- Independent and non-conformist thinking,
- Preference for solitary academic activities for long periods.

Types of Intelligence Tests

(1) Individual or Group Tests

Individual Tests

- Can be administered to one person at a time
- Require test administrator to establish a rapport with the subject & be sensitive to her/his feelings, moods & expressions during test
- Allow people to answer orally, or in a written form

Group Tests

- Can be administered to several people simultaneously
- Do not allow an opportunity to be familiar with the subjects' feelings, thus no rapport formation
- Generally seek answers in a multiple-choice format.

② Verbal, Non-Verbal or Performance Tests

- Verbal Tests require subjects to ~~show~~ give verbal responses either orally or in a written form. Thus, these tests can be administered to only literate people.
- Non-Verbal tests use pictures or illustrations as test items.
eg- Raven's Progressive Matrices (RPM)
- Performance Tests require subjects to manipulate objects and other materials to perform a task. Written language is not necessary. They can be easily administered to people of different cultures.
eg- Kohs' Block Design Test

③ Culture-Fair or Culture-Biased Tests

- Culture-Fair Tests or culturally appropriate are the ones which do not discriminate against individuals belonging to different cultures. Assess experiences common to all cultures. Non-Verbal & Performance tests help reduce the cultural bias.
- Culture-Biased Tests are tests that show a bias ~~to~~ the culture in which they are developed, and assess experiences common to 1 specific culture.

Intelligence Testing in India

- S.M. Mohsin made a pioneering attempt in constructing an intelligence test in Hindi.
- In 1930s, C.H. Rice attempted to standardise Binet's test in Urdu and Punjabi.
- Mahalanobis attempted to standardise Binet's test in Bengali.

- Long & Mehta prepared a Mental Measurement HandBook listing out 103 tests of Intelligence in India that were available in various languages.
- Bhatia's Battery of Performance Tests is quite popular.
- The National Library of Educational and Psychological Tests (NLEPT) at the NCERT has documented Indian tests.

● Culture and Intelligence

- The cultural environment provides a context for intelligence to develop.
- Vygotsky, a Russian psychologist, has argued that culture provides a social context in which people live, grow, and understand the world around them.

eg.

In less technologically developed societies, social & emotional skills in relating to people are valued.

While, in technologically advanced societies, personal achievement founded on abilities of reasoning and judgement is considered to represent intelligence.

- "Culture is a collective system of customs, beliefs, attitudes, and achievements in art and literature."
- Sternberg's notion of contextual or practical intelligence implies that intelligence is a product of culture.

- Vygotsky also believed that cultures, like individuals have a life of their own; they grow and change, and in the process specify what will be the end-product of successful intellectual development.
- Technologically advanced societies promote a type of behaviour called "Technological Intelligence". In these societies, persons are well-versed in skills of attention, observation, analysis, performance, speed and achievement orientation.
- Non-Western societies value self-reflection and collectivistic orientation as opposed to personal achievement & individualistic orientation in western societies.

Intelligence in the Indian Tradition

- Intelligence in the Indian Tradition can be termed as 'Integral intelligence', which gives emphasis on connectivity with the social and world environment.
- 'Buddhi', according to J. P. Das, includes such skills as mental effort, determined action, feelings, and opinions along with cognitive competence such as knowledge, discrimination, and understanding.
- 'Buddhi' is the knowledge of one's own self based on conscience, will and desire.

The following competencies are identified as facets of intelligence in the Indian tradition :

- Cognitive Capacity - sensitivity to context, understanding discrimination, problem solving, & effective communication
- Social Competence - respect for social order, commitment to elders, the young and the needy, concern about others, recognising others' perspective)
- Emotional Competence - self-regulation & self-monitoring of emotions, honesty, politeness, good conduct, and self-evaluation.
- Entrepreneurial Competence - commitment, persistence, patience, hard work, vigilance, and goal directed behaviours.

• Emotional Intelligence

- Emotional Intelligence is a set of skills that underlie accurate appraisal, expression, and regulation of emotions. It's the feeling side of intelligence.

- The concept was first introduced by Salovey and Mayer who considered emotional intelligence as "the ability to monitor one's own & others emotions, to discriminate among them, and to use the information to guide one's thinking and actions."

- Emotional Quotient (EQ) is used to express emotional intelligence.

- Characteristics of Emotionally Intelligent Persons :

- Perceive & be sensitive to your feelings & emotions
- Perceive & be sensitive to various types of emotions in others by noting their body language, voice and tone, & facial expressions
- Relate your emotions to your thoughts so that you take them into account while solving problems & taking decisions.
- Understand the powerful influence of the nature of emotions
- Control & regulate your emotions & their expressions while dealing with self & others to achieve harmony and peace.

• Special Abilities

Aptitude : Nature and Measurement

- Aptitude refers to special abilities in a particular field of activity. It is a combination of characteristics that indicates an individual's capacity to acquire some specific knowledge or skill after training.
- In order to be successful in a particular field, a person must have both aptitude and interest.
- Interest is a preference for a particular activity; aptitude is the potentiality to perform that activity.
- Aptitude tests are available in 2 forms :
 - (a) Independent (specialised) aptitude tests
eg- Clerical aptitude, Mechanical aptitude, Numerical aptitude, & Typing aptitude
 - (b) Multiple (generalised) aptitude tests
eg- Differential Aptitude Tests (DAT), General Aptitude Test Battery (GATB), Armed Services Vocational Aptitude Battery (ASVAB).

- DAT is most commonly used in educational settings. It consists of 8 independent subtests :

- | | |
|--------------------------------|--------------------------|
| (i) Verbal Reasoning | (v) Mechanical Reasoning |
| (ii) Numerical Reasoning | (vi) Space Relations |
| (iii) Abstract Reasoning | (vii) Spelling |
| (iv) Clerical Speed & Accuracy | (viii) Language Usage |

J. M. Qjba has developed an Indian Adaptation of DAT.

• Creativity

- Research suggests that children begin to develop their imagination during the early years of childhood but they express Creativity mostly through physical activities & in non-verbal ways.
- Creativity is determined by both heredity & environment.
- Limits of the creative potential are set by heredity, environmental factors stimulate the development of creativity.

Creativity and Intelligence

- A person who has the ability to learn faster and reproduce accurately may be considered intelligent more than creative unless s/he devises new ways of learning and doing.
- Teerman, in the 1920s, found that persons with high IQ were not necessarily creative. Intelligence, therefore, by itself does not ensure creativity.
- However, all creative work requires a certain level of intelligence, but beyond that intelligence does not correlate well with creativity.

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- A general feature of most of the creativity tests is that they are open-ended. They permit the person to think of different answers to the questions or problems in terms of her/his experiences.

There is a freedom to use one's imagination & express it in original ways.

MAKE SURE YOU ALSO WATCH THE VIDEO TO UNDERSTAND !

REVISE WELL !

ALL THE BEST 😊

Sakshi

Mind Review